

THE ENGINEERING CHALLENGE

Structuring the global artificial intelligence landscape requires real-time mining of highly heterogeneous, volatile sources. Traditional extraction methods suffer from extreme DOM variations, payload sizing bottlenecks, transient network issues, API rate limiting, and duplication, causing rapid schema drift and cascading storage failures.

THE AIIP ARCHITECTURE

AIIP acts as the high-throughput, unified data-gathering layer for GraphOne. It decouples high-concurrency transport from processing and multi-format storage. Built entirely with non-blocking async operations, it enforces strict schema validation and applies algorithmic record-level deduplication prior to multi-format distribution.

TOTAL VERIFIED RECORDS

8,279

● Live Production Count

CONFIGURED SOURCES

21 Source Engines

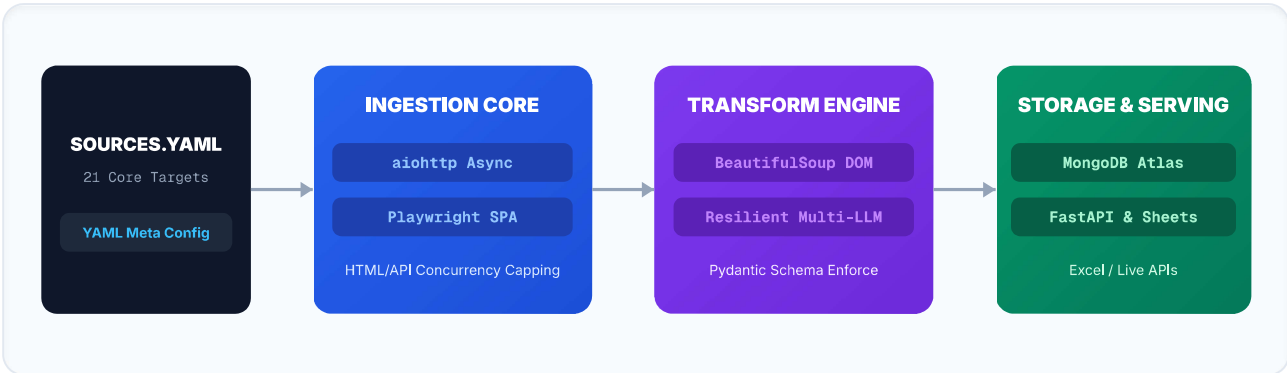
src/config

ACTIVE OUTPUT CHANNELS

6 Unified Formats

CSV/XLSX/Web

CORE SYSTEM TOPOGRAPHY & BLOCK DATA-FLOW

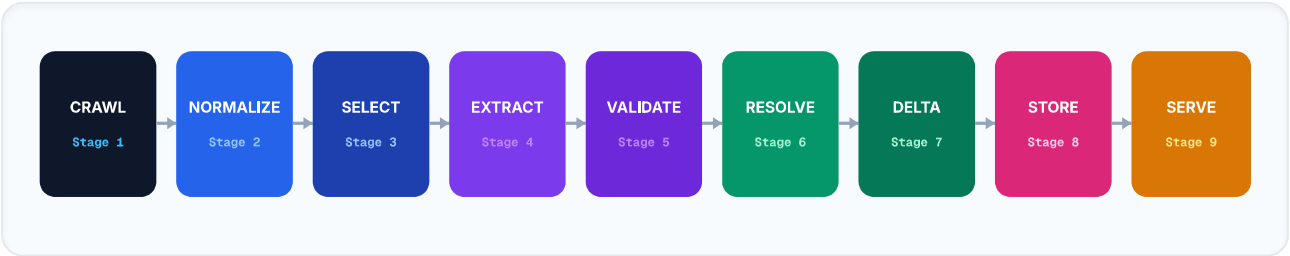


DATA SOURCES & VERIFIED PRODUCTION YIELD

DOMAIN CATEGORY	PRIMARY SOURCES & METHODS	BASELINE TARGET	VERIFIED DB COUNT	PRODUCTION COMPLIANCE STATUS
● Startups	YC Companies REST API direct fetch	1,000+	5,754	Exceeds (+475%)
● Products	GitHub Search & Trending AI Scrapers	1,000+	1,103	Exceeds (+10%)
● Research	arXiv API Atom / XML response feeds	1,000+	1,097	Exceeds (+9.7%)
● Jobs	RemoteOK Table Scraping / WWR LLM Extractor	50+	149	Exceeds (+198%)
● News	TechCrunch, Wired, Google AI Feeds (Inference)	50+	176	Exceeds (+252%)

End-to-End Ingestion Flow

MODULAR STAGE SEQUENCE (`SRC/MAIN.PY`)



1

ASYNC CRAWLER LAYER

Uses high-concurrency `aiohttp` and non-blocking sockets. For SPAs, a chromium instance via `Playwright` handles asynchronous hydration dynamically.

5

SCHEMA VALIDATION

Enforces type validation via strict `Pydantic v2` data models, eliminating bad schemas and button artifacts before saving files.

2

DOM NORMALIZATION

Strips raw clutter (e.g., scripts, styles, inline svg graphics, and nav boilerplate) to significantly reduce prompt token overhead by removing boilerplate HTML.

6

ENTITY RESOLUTION

Applies `RapidFuzz` matching algorithms against seed databases to cleanly merge name records (e.g. "Open AI" and "OpenAI Inc").

3

EXTRACTION ROUTING

Evaluating the targeted URL configuration routes processing tasks instantly to direct REST endpoints, embedded JSON-LD payloads, or the deep AI engine.

7

KNOWLEDGE DELTA DIFFING

Builds persistent file signatures via custom calculated hashes. Matching hash signatures bypass write requests, optimizing execution cycles.

4

HYBRID EXTRACTION

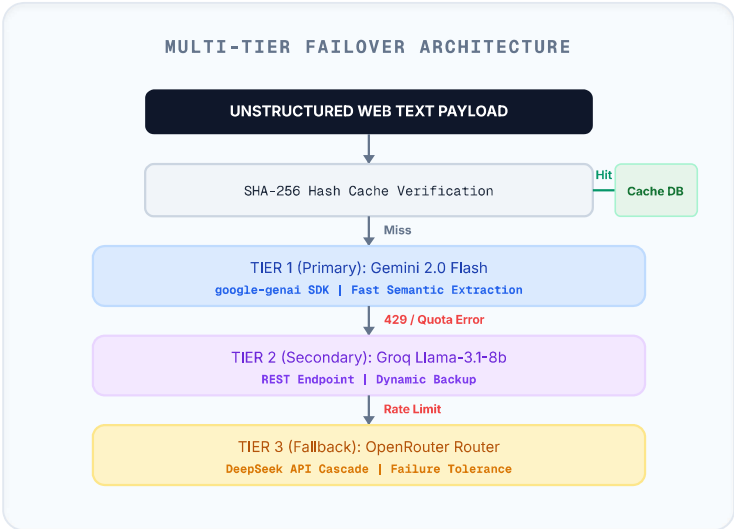
Uses semantic extraction models when scraping unstructured layouts, whereas structured tables are parsed via highly reliable DOM paths to keep processing clean.

8

UNIFIED EXPORTER

Aggregates output into single DataFrames, pushing clean data structures instantly into MongoDB, Excel workbook files, and synced Google Sheets sheets.

Core Subsystem Architecture



MULTI-TIER FAILOVER ENGINE

Extraction routing adapts automatically during transient failure events. Built-in error handling monitors model responses, switching downstream engines in real time.

TOKEN OVERRUN STRATEGY

When encountering an HTTP 413 error (Payload Too Large), AIIP triggers semantic chunking using a strict threshold limit:

```
_MAX_CHUNK_CHARS = 3500
```

RESOLUTION ENGINE & SCHEMA FINGERPRINTING

ENTITY RESOLUTION LOOP

RapidFuzz

Matches incoming variations dynamically using raw token ratio scoring against Y Combinator seed lists.

"Open AI, Inc."

> Match ≥ 88%

"OpenAI"

DELTA FINGERPRINT CONTROL

SHA-256

Generates a composite content signature to identify modified fields and execute conditional database upserts.

Fingerprint = SHA256(Name + Domain + ContentHash)

MATCH: Skip Write IOPS

DIFF: Update History

UNIFIED TARGET EXPORTER ARCHITECTURE

Single Source of Truth Pattern: Raw datasets are compiled into a normalized intermediate DataFrame before distribution. This pattern prevents schema mismatch across exports.

Mongo Atlas

6 CSVs

Sheets Sync

Deployment & Scalability Specification

FASTAPI PRODUCTION DELIVERY

The FastAPI application is deployed live on Render Cloud Services. Endpoints feature memory-efficient pagination and regex query parameters to minimize server load.

ACTIVE ENDPOINTS

GET

/dataset

GET

/download/excel

GET

/health

aip-api.onrender.com/docs

Swagger UI v3.0

GET

/startups

Retrieve YC records

offset/limit

GET

/download/excel

Stream raw excel data

app/excel

FUTURE ENTERPRISE SCALE-OUT ARCHITECTURE (TARGET CAPACITY: 500K+)



KEY ARCHITECTURAL DECISIONS

●

DETERMINISTIC PARSERS VS LLM INFERENCE

Using LLMs for structured DOM components adds cost and latency. Dedicated `BeautifulSoup4` scrapers are maintained for stable, tabular endpoints (e.g. RemoteOK tables). Unstructured assets are routed selectively to our LLM cluster.

Result: Drastically reduces API token burn and limits rate-limit blocks

●

UNIFIED DATAFRAME PIPELINE SPEC

Direct exports to multiple databases (MongoDB, Google Sheets, flat files) often result in schema misalignment. Extracted elements pass through a single, central DataFrame mapping.

Result: Guarantees schema parity across database targets